MACHINE LEARNING AND AI FOR ANALYTICAL CHEMISTRY

Part 2: Data origin, rights, and quality

M2 Sciences analytiques - 2 h lectures

WHAT THIS PART COVERS

Roadmap for Part 2

1

Data for training

Where data come from, their characteristics, licensing of code and data, and intellectual property

2

Levels and quality

Levels of information, how AI hides the base, poisoning the well, FAIR stewardship, model documentation

DATA FOR TRAINING

Where the data come from

Your own lab

Instruments you control: known conditions, but limited coverage.

Public repositories

Depositions and libraries: broad coverage, varied quality and licences.

Literature

Papers and supplementary data: useful but often poorly structured.

Simulated / augmented

Generated or perturbed data: fills gaps, risks unrealistic patterns.

Example: a retention-time model trained on your lab, on ProteomeXchange/PRIDE, or on the NIST library covers different compounds and conditions.

Characteristics to check first

Coverage & balance

Are all classes or ranges represented, or is one dominant?

Label quality

Are targets correct? Pipeline labels carry the pipeline's errors.

Batch & instrument effects

One instrument, operator or season can leak into the model.

Metadata

Acquisition conditions must be recorded, or data cannot be reused.

Example: MS/MS spectra from several instruments differ systematically; a model may learn the instrument, not the chemistry.

Licensing: code and data are separate

	Permissive / open	Restrictive
Code	MIT, Apache-2.0 (broad reuse)	GPL (share-alike derivatives)
Data	CC0, CC-BY (use, attribute)	CC-BY-NC (no commercial use)

Credentials: never hard-code API tokens; use environment variables or a config file kept out of version control.

Example: a fragmentation model trained on CC-BY-NC spectra cannot be shipped commercially; the data licence propagates to the model.

Intellectual property and ownership

Who owns the data?

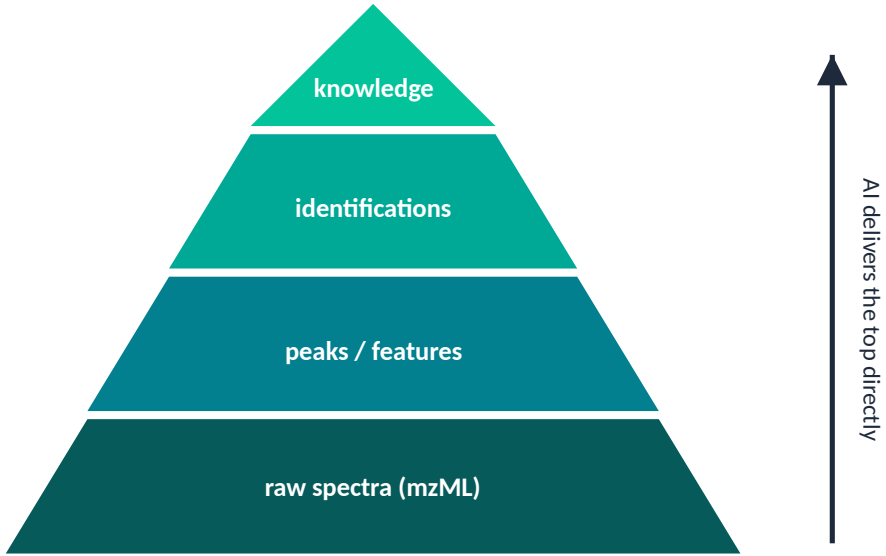
Beyond the licence, ask who is legally entitled to the data. Instrument data produced under a service contract, a collaboration, or industrial funding may belong to a party other than the person who ran the sample. Clarify ownership and consent before using data to train a model that will be shared or published.

Example: raw LC-MS files acquired for a paying client are usually the client's property; training a public model on them needs permission.

Levels and quality of information

From raw signal to knowledge - and who keeps the base trustworthy

Levels of information, and where AI looks



information is lost at each step upward

Data exist from raw signal to interpreted knowledge; each level is derived from the one below, losing detail.

AI tools increasingly deliver the top directly, which pushes the foundational data out of view.

If the model is wrong at the bottom, the error is invisible at the top, where the user looks.

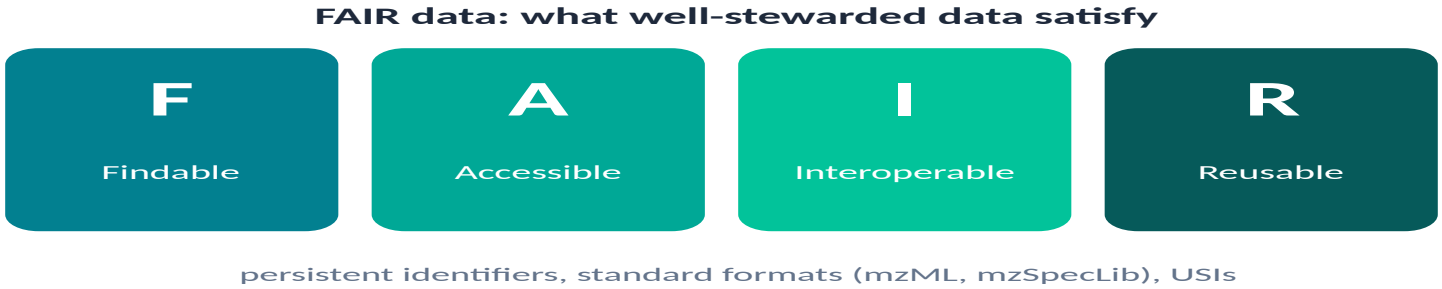
Data quality and "poisoning the well"

Shared foundational data are a single point of failure.

Because models and downstream users all depend on the same source data, an error at the source propagates everywhere. A mislabelled library entry is copied into every model trained on it and every analysis that trusts it, and it is hard to trace because the source looks authoritative.

Example: one misannotated reference spectrum makes every tool that trusts the library repeat the same wrong identification.

Who maintains the base? FAIR stewardship



Sustained curation is a real cost carried by repositories, consortia and standards bodies. **FAIR** data are Findable, Accessible, Interoperable and Reusable; persistent identifiers and standard formats make this practical.

Example: ProteomeXchange/PRIDE give stable accessions; mzML/mzSpecLib give interoperability; a USI points to one exact spectrum.

Documenting models and datasets

Model cards

Structured documentation that travels with a model: intended use, training data and its licence, evaluation, and known limitations. Version both data and models for reproducibility.

Why it matters

A future user must know what a model learned from, what it is for, and where it breaks, before relying on it. Community resources such as ProteomicsML share documented models and datasets.

Example: a published retention model with a model card stating training set, licence, column and error lets another lab judge fit before use.

KEY TAKEAWAYS

- A model is only as good, and as legal, as the data behind it.
- Check origin, balance, label quality, batch effects and metadata first.
- Code and data carry separate licences; data licences propagate to the model.
- AI can hide the foundational data, so errors at the base go unseen.
- FAIR stewardship and model cards keep the shared base trustworthy.